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Green preparation of branched biolubricant by chemically modifying waste cooking oil with lipase and ionic liquid

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ABSTRACT

To reduce the environment pollution from biolubricant synthesis, in this study, we developed a green and efficient strategy for the preparation of octylated branched biolubricant from waste cooking oil (WCO): (1) WCO was first hydrolyzed by a lipase (*Candida* sp. *99–125*) to produce unsaturated fatty acids (UFAs), the latter was subsequently concentrated by urea complexation; (2) Next, we synthesized new esters using the UFAs as substrate via esterification with 2-ethylhexanol employing lipase (Novozym 435) and the new esters products were further epoxidated; (3) After that, the epoxy group was attacked by a low-cost nucleophilic reagent, octanoic acid, to prepare octylated branched biolubricant using an recyclable ionic liquid, [HMIm][PF_6], as catalyst. After three recycling times, [HMIm][PF_6] remained catalytic activity with a conversion yield of 81.22% and selectivity of 82.10%. The obtained product showed excellent lubricant properties, such as low pour point of −61 °C, high viscosity index of 149 and high thermal-oxidation stability (onset temperature of 326.35 °C and oxidative onset temperature of 312.06 °C). The high frequency reciprocating rig (HFRR) measurement demonstrated that lubrication performance (friction coefficient of 0.09 and wear spot diameter of 203 μm) was much better than that of mineral based lubricant with the same viscosity grade. Therefore, this study provided a critical synthetic strategy for preparing high-performance biolubricants from WCO.

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1. Introduction

In 2017, the global demand of lubricants automotive reached approximately 39 million tons, of which the consumption of automotive lubricants was more than 60% (Hamdan et al., 2018). As the increasing of vehicles and the degree of mechanical industrialization, the consumption of lubricants will continue to increase with a rate of 2% annually (Singh et al., 2017). Moreover, the non-renewable petroleum-based lubricants with poor environmental compatibility accounted for 85%–90% of total consumption. Their large use has resulted in greenhouse gas emission and water pollution (Soufi et al., 2019). Therefore, biolubricants as the alternatives to petroleum-based lubricants, are attracting growing attention (McNutt and He, 2016).

Biolubricants have many advantages in lubrication performance when compared with petroleum-based lubricants, such as low volatility, wide viscosity range, high temperature tolerance, good viscosity-temperature property, sustainability, and environmental friendliness (Gorla et al., 2014; Sarno et al., 2019; Romsdahl et al., 2019). Besides, their raw materials are originated from animal and vegetable oils. Nevertheless, using edible vegetable oils, such as soybean oil and rapeseed oil, to produce biolubricants will lead to a conflict between energy sustainability and food safety (Kousaalya et al., 2018). Simultaneously, the non-edible oils such as jatropha oil and castor oil are usually originated from perennial plants. The limitation of planting regions and climates fails to meet the demand of biolubricants in the future (Mansir et al., 2018). On the other hand, about 15–18 million tons of WCO around the world is discharged into water or soil randomly per year (Mansir et al., 2018; Li and Wang, 2015). In China, about 5 million tons of WCO is produced every year, 40–60% is illegally returned to kitchen (Zhang et al., 2012). Therefore, employing WCO to produce high value

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List of abbreviations

[HMIm][PF_6]	1-Hexyl-3-methylimidazolium hexafluorophosphate
AV	Acid value
COF	Coefficient of friction
DTG	Differential thermogravimetry
FP	Flash point
FT-IR	Fourier transform infrared spectroscopy
GC	Gas chromatography
HRFF	High frequency reciprocating rig
IV	Iodine value
NMR	Nuclear magnetic resonance
PP	Pour point
RPVOT	Rotating pressure vessel oxidation test
TG	Thermogravimetry
UFAs	Unsaturated fatty acids
VI	Viscosity index
WCO	Waste cooking oil
WCOME	Waste cooking oil methyl esters
WSD	Wear spot diameter

chemicals, as showed in Table 1, such as biofuels (biodiesels or aviation fuels), polymeric materials or bioplasticizers and biolubricants will display a significant potential in the future. Most importantly, the biggest obstacle of the biolubricants development is the high raw materials cost, about 70–80% of the total cost, while the cost of WCO is 2–3 times lower than that of animal and vegetable oils (Dang et al., 2016). Using biolubricants with good performance and economic values produced from WCO can reduce the damage to human health, alleviate environmental burden (Rajaeifar et al., 2017).

The fatty acids of WCO are mainly composed of oleic acid, polyunsaturated fatty acids, palmitic acid, and stearic acid (Liu et al., 2014). Therefore, it is possible to prepare biolubricants with better low-temperature performances and thermal-oxidation stability by chemically modifying the unsaturated fatty acids in WCO. Epoxidation of C=C double bond, hydroxylation of epoxy groups and further acylation are the noteworthy method for the chemical modification (Salimon et al., 2012a). Borugadda and Goud (2016) used acidic ion exchange resin (IR-120) to modify WCO methyl esters (WCOME) to obtain epoxide via in-situ epoxidation. Produced biolubricant basestock showed a high viscosity and thermal-oxidation stability. However, by introducing epoxy groups to increase intermolecular force and accelerate crystallization resulted in a poor low-temperature performance. Raghunanan and Narine (2016) prepared hydroxylated and propionylated branched derivatives from 1,6-(oleoyloxy) hexyloleate diester through epoxidation and acylated ring opening reactions. The viscosity of branched biolubricants was 10 times higher than that of initial diesters and the whole glass transitions bellowed −65 °C. Borugadda and Goud (2019) reported a synthetic method of branched biolubricants by hydroxylation (introducing alcohols with different chain lengths) and subsequent acylation (caproic anhydride) of epoxidized WCO and WCOME. The obtained products showed good thermal-oxidation stability, low-temperature performance, lubricity, and biodegradability. Meanwhile, to meet the requirement of lubricants in extreme environment, the polyacyl-substituted and branched biolubricants with high viscosity and low pour point (PP) were investigated by attacking epoxy groups with various nucleophilic reagents, such as acetic anhydride, propionic anhydride, butyric anhydride and hexanoic anhydride. However, the supported noble metal catalysts were very expensive and the reaction selectivity of the acidic inorganic acids and resins was relatively low during the preparation of branched biolubricants.

Comparing with these catalysts, lipases and ionic liquids are characterized in high catalytic efficiency, high specificity, good recyclability in chemical structure modification of UFAs and esters

Table 1
Bioproducts obtained from waste cooking oil (WCO) and performance evaluation.

Raw materials	Products	Findings	Ref.
WCO	Biodiesel	The cheaper CaO/SiO_2 as a heterogeneous catalyst, the yield of biodiesel was 91%, and the mechanism and kinetics of transesterification were studied.	Putra et al. (2018)
WCO	Biodiesel	Exergoeconomic and exergoenvironmental approaches were applied to produce biodiesel, per unit of exergy were respectively found to be 2.23 USD/GJ and 1.32 mPts/GJ, the yield of biodiesel was higher than 96.5% in a piezoelectric ultrasonic reactor using a coupled scheme of ANFIS and ALIFMO-NSGA-II.	Aghbashlo et al. (2018), 2019
WCO and brown grease	Biodiesel	The best performance with 100% conversion was catalyzed by Candida rugosa lipase nanoparticles conjugated to magnetic nanoparticles, the stability was sustained even after five consequent biodiesel batch.	Sharma et al. (2019)
WCO	Biodiesel	Biodiesel could potentially decrease burdens of ecosystem, human health and energy through life cycle assessment.	Rajaeifar et al. (2017)
WCO	Biodiesel	The comprehensive performance of biodiesel purified by Span 80 microemulsion was close to that of biodiesel/diesel (V/V, 5/5) blend.	Leng et al. (2019)
Fatty acid methyl esters from WCO	Hydrocarbon biofuels or aviation fuels	The hydrodeoxygenation yield could reach 97% catalyzed by sulfur-free Ni–Mo/γ-Al_2O_3.	Wang et al. (2017)
Epoxidized WCO	Bio-polyol for bioplasticizers	Bio-polyols based on waste oils with various characteristics can be obtained using different types of catalysts, the tetrafluoroboric acid solution in water reduced reaction temperature and reaction time.	Kurańska et al. (2019)
WCO	TMP triester for biolubricants	The TMP triester conversion was obtained at 97% catalyzed by CaO derived from waste cockle shell.	Ghafar et al. (2019)
Waste cooking oil methyl esters (WCOME)	Epoxidized WCOME for biolubricants	Structurally modified epoxidized WCOME had better viscosity and thermal-oxidative stability.	Borugadda and Goud (2016)
WCO	Three epoxidized branched esters for biolubricants	After epoxidation and two-step transesterification, branched chain was introduced into the molecular structure, the low temperature performance, antioxidation and thermostability, lubrication performance of products were much better.	Li and Wang (2015)

(Zhang et al., 2020). So we developed a green and efficient strategy for synthesis of branched biolubricant from WCO using lipases and ionic liquid as catalysts. The specific routes were as follows: 1) WCO was hydrolyzed by a lipase (*Candida* sp. *99–125)*. 2) The produced UFAs were collected from the hydrolysates by the urea complexation separation. 3) Another lipase (Novozym 435) was used to catalyze the esterification of UFAs with 2-ethylhexanol and the new esters products were further epoxidated. 4) The cheaper nucleophilic reagent (octanoic acid) was used for nucleophilic disubstitution of the epoxy groups under the catalysis of ionic liquid to obtain octylated branched biolubricant. Finally, the low-temperature performance, thermal-oxidation stability, and lubrication performance of branched biolubricant were evaluated.

2. Experimental section

2.1. Materials

The WCO with an iodine value (IV) of 98.14 gI_2/100 g and an acid value (AV) of 67.82 mgKOH/g was provided by Shanghai Lvming Environmental Technology Co., Ltd, China. Urea and 30 wt% hydrogen peroxide were purchased from Beijing Chemical Factory, China. Lipase (*Candida* sp. *99–125*) with a hydrolase activity of 20,000 U/g was provided by Beijing CTA New Century Biotechnology Co., Ltd. Lipase (Novozym 435) was obtained from Novozymes (China). Octoic acid and 2-ethylhexanol (AR, purity >98%) were purchased from Aladdin reagent (Shanghai) Co., Ltd. 1-Hexyl-3-methylimidazolium hexafluorophosphate ([HMIm][PF_6]) was obtained from Lanzhou Institute of Chemical Physics, Chinese Academy of Sciences. Other reagents were purchased from Fuchen (Tianjin) Chemical Reagent Co., Ltd, and all reagents are analytical grade except for special marking.

2.2. Analytical methods

2.2.1. Characterization of products

The chemical structure of all samples were characterized by ^1H NMR and ^{13}C NMR using Agilent 600 MHz DD2 NMR spectrometer (Agilent Technologies Inc., USA). Fourier transform infrared spectroscopy (FT-IR) analysis were performed on a Nicolet 6700 Thermo Electron Corporation (USA) equipped with a DTGS KBr detector and a KBr beam splitter according to a liquid membrane method. The scanning range was between 4000 cm^{-1} and 400 cm^{-1} at a spectral resolution of 4 cm^{-1}.

2.2.2. Gas chromatography analysis

The samples in synthetic process were analyzed by gas chromatography (GC) on DB-1ht column using a Thermo Scientific TRACE1300 instrument. The column specification was 30 m × 0.250 mm × 0.10 μm (Agilent Technologies Inc., USA). The temperature of GC column gradually increased from the initial 120–180 °C with an increment of 30 °C/min, then increased to 320 °C with an increment of 45 °C/min and maintained for 25 min. The temperature of detector was set at 320 °C. 10 uL sample was dissolved in n-hexane (which could dissolve all compounds) for GC analysis. Each sample was run in triplicape and the average values were reported.

2.2.3. Determination of physicochemical properties

The AV of all samples was obtained according to the method AOCS Te 1a-64. The kinematic viscosities at 40 °C and 100 °C were measured using a Kinematic viscosity Instrument (SYD-265H, Wuhan Gelaimo Testing Equipment Co., Ltd) according to ASTM D445. The viscosity index (VI) was calculated according to ASTM D2270. The PP was measured on an Automatic Pour Point Apparatus (SYD-510Z-2, Shanghai Wang Xu Electric Co., Ltd) according to ASTM D5949 (2011a). The flash point (FP) was measured using an Cleveland Open Fash Point Tester (SYD-3536, Shanghai Wang Xu Electric Co., Ltd) according to ASTM D93. Furthermore, the IV and epoxy oxygen group contents of the samples were measured by titration according to the method AOCS official method Tg 1a-64 and Cd 9–57, respectively. The conversion yield of oxirane and theoretical oxirane oxygen content (OO_{th}) were obtained according to previous methods (Lu et al., 2010). Above measurements were run in triplicape and the average values were reported. And rheological behaviors of WCO and octylated branched biolubricant were measured by a Rotational Rheometer (AR2000CS) under various shear rates (0–500 S^{-1}) and different temperatures (30, 40, 50, 60, 80 and 100 °C).

Thermogravimetry (TG) analysis of WCO and octylated branched biolubricant was carried out using PerkinElmer thermogravimetric differential thermal analyzer (TGA5500). The temperature ranged from 30 °C to 600 °C at a constant heating rate of 10 °C/min. Heating was proceeded in N_2 atmosphere at a flow rate of 40 mL/min to obtain the thermal stability or in atmosphere at a flow rate of 40 mL/min to obtain the oxidative stability. Meanwhile, oxidation stability was assessed using an Automatic Lubricant Oxidation Stability Tester (RP-0193-1, Hebi Ruipu Instrument Co., Ltd) for Rotating pressure vessel oxidation test (RPVOT) according to ASTM D2272-02 at 150 °C.

The tribological test was carried out according to ASTM D6079-04 using HFRR. The metal sheet was put into the oil tank containing test lubricants at 60 °C. And the metal ball loading 600 g continuously moved back and forth on the surface of the metal sheet at 50 Hz for 75 min. The contact surface between the metal ball and the metal sheet completely was submerged in test lubricants. Wear spot diameter (WSD) on the surface of metal ball were measured by an electron microscope, and coefficient of friction (COF) was obtained from a Surface Roughness On-line Measurement System.

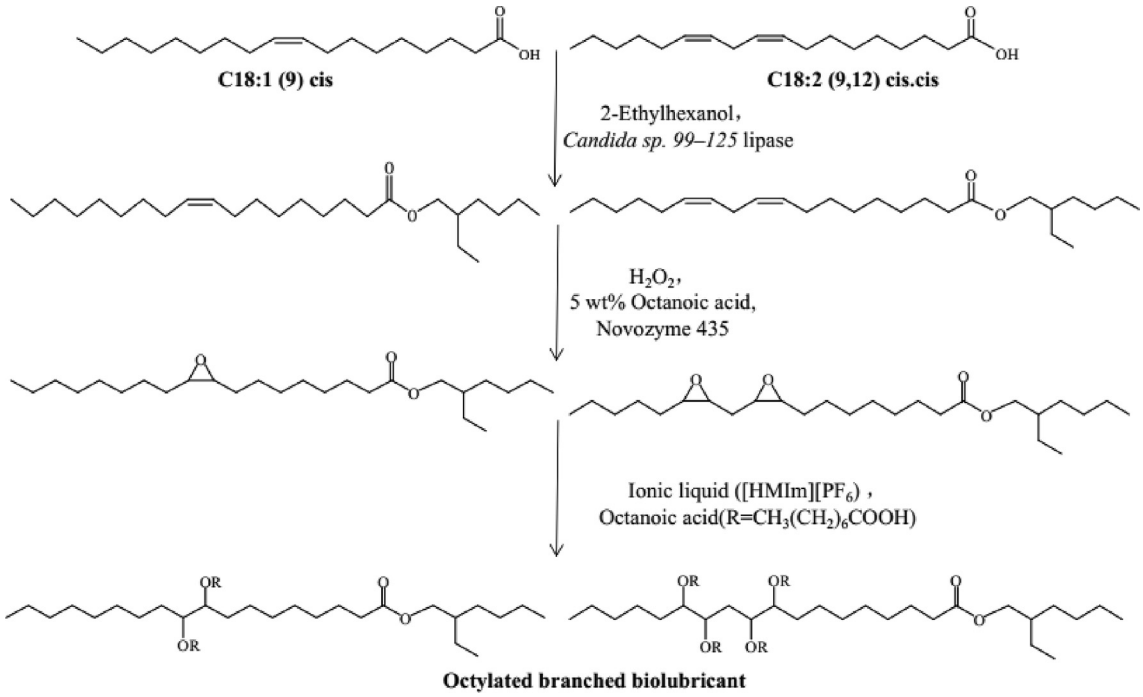
2.3. Synthesis of branched biolubricant

2.3.1. Hydrolysis of waste cooking oil and removal of saturated fatty acids

The enzymatic hydrolysis of WCO and the collection of UFAs via urea complexation separation were improved according to previous research (Wang et al., 2014). The hydrolysis system contained 400 g WCO, 480 g phosphoric acid buffer (pH 7.5), and lipase (*Candida* sp. *99–125*) of 600 U/g oil. The reaction was carried out in a 2 L three neck flask at 35 °C for 72 h with a stirring speed of 350 rpm. 2.0 mL sample was withdrew for 3 h interval and subsequently centrifugated. The upper oil phase was dried by anhydrous sodium sulfate for acid value titration. The yield of hydrolysis was calculated by the equation (1). After reaction, the obtained oil phase by centrifugation and drying was distilled by short-range molecular distillation at 140 °C in low vacuum to obtain free fatty acids.

$$Yield\ of\ hydrolysis\ (\%) = \frac{AV\ of\ theoretical - AV\ of\ sample}{AV\ of\ theoretical} \quad (1)$$

Distilled fatty acids, urea, and methanol (1:2:5, w/w) were dropped into the three neck flask equipped with a reflux and mechanical agitation device stirring at 500 rpm and 50 °C for 1 h, then cooled down to 0 °C for 2 h. The solid-liquid mixtures was separated through a Buchner funnel and the filtered liquid was evaporated in a rotary evaporator. After adding 2–3 times hot water washing, the upper oil phase was dried to obtain UFAs. The fatty acids yield of non-urea complexes was calculated by the equation (2).



Scheme 1. The specific scheme for the preparation of octylated branched biolubricant.

$$\text{Yield of non} - \text{urea complexes}\,(\%) = \frac{\text{Weight of non} - \text{complexation fatty acids}}{\text{Weight of fatty acids brfore complexation}} \quad (2)$$

2.3.2. Synthesis of 2-ethylhexyl ester and its epoxidation

The specific synthesis path of branched biolubricant is shown in Scheme 1. The esterification of UFAs with 2-ethylhexanol was carried out in a 250 mL three neck flask. Fatty acids, alcohol (molar ratio 1:1.1), 5 wt% lipase based on substrates (*Candida* sp. *99–125*) were mixed at 350 rpm and 50 °C for 12 h. The produced water during the reaction was timely removed under low pressure by a vacuum pump to promote the accumulation of esters. 1.0 mL sample was withdrew at interval of 1 h and centrifugated for acid value titration. The final esterification yield calculated by acid value titration was 98.78%. Finally, the obtained oil phase after centrifugation and drying was distilled by short-range molecular distillation at 80 °C in low vacuum to remove excess 2-ethylhexanol and unreacted acid. The AV of 2-ethylhexyl esters was 0.49 mgNaOH/g.

According to adapted methodology described by Zhang et al. (2017), 30% H_2O_2 was added into the reaction system through an injection pump to prevent lipase deactivation during epoxidation of 2-ethylhexyl esters. 2-Ethylhexyl esters, 5 wt% octanoic acid based on esters (Forming peroxyacid to enhance epoxidation of esters reported by Lu et al. (2010) and Chua et al. (2012)), and 5 wt% lipase (Novozym 435) based on substrates were mixed at 350 rpm and 50 °C for 24 h in a 250 mL three neck flask. At the same time, 30% H_2O_2 (N_{H2O2}:$N_{C=C}$=1.2:1) was uniformly added into the reaction system through an injection pump within 24 h. 3.0 mL sample was withdrew at interval of 1 h and centrifugated, and the upper oil phase was dried by anhydrous sodium sulfate for determination of IV and epoxy value. Finally, lipase and water were removed by centrifugation and the upper oil phase was dried for next step reaction. The oxirane content of epoxy 2-ethylhexyl esters and the epoxy conversion yield were 5.37% and 89.05%, respectively.

2.3.3. Synthesis of branched biolubricant

For improving the low-temperature performance of epoxy 2-ethylhexyl esters, the epoxy group was cleaved by a nucleophilic reagent (octanoic acid) under the catalysis of [HMIm][PF_6] to form a dioctylated branched biolubricant. Epoxy 2-ethylhexyl esters to octoic acid (molar ratio 1:5) and 5 wt% [HMIm][PF_6] based on esters were mixed and stirred at 500 rpm and 150 °C for 24 h in a 250 mL three neck flask. During the reaction, 1.0 mL the upper oil phase of each sample after centrifugation completely solubled in n-hexane for GC detection. The conversion yield and selectivity were calculated by the equations (3) and (4), respectively, according to an area normalization method. Reaction mixtures were cooled down to room temperature and centrifuged ulteriorly. The upper oil phase was distilled by short-range molecular distillation at 175 °C in low vacuum to obtain octylated branched biolubricant. Ionic liquid in bottom layer was washed with n-hexane for about 3 times to remove the residual reactants and products, and subsequently dried in an oven at 100 °C for 12 h.

$$\text{Conversion}\ (\%) = \frac{S_P}{S_P + S_E} \times 100 \quad (3)$$

$$\text{Selectivity}\ (\%) = \frac{S_{TP}}{S_P} \times 100 \quad (4)$$

S_P: peak area of all products; S_E: peak area of substrate ester; S_{TP}: peak area of target product.

3. Results and discussion

3.1. Synthesis of branched biolubricant

3.1.1. Hydrolysis of waste cooking oil and urea complexation

WCO was enzymatically hydrolyzed for 72 h and subsequently centrifuged and dried to collect fatty acids. The AV was detected to be 184.92 mgNaOH/g while the theoretical AV was 204.13 mgNaOH/g. According to the equation (1), the hydrolysis yield was

90.57%. The accumulation of fatty acids in the later stage of hydrolysis caused decrease of pH value and the deactivation of partial enzyme. Besides, the increased concentration of the saturated fatty acids made the mass transfer become more difficult. The results of hydrolysis and urea complexation are shown in Table 2. The saturated fatty acids and urea formed complex crystallization were removed by filtration. The contents of UFAs in the filtrate was 98.04%.

3.1.2. Synthesis of octylated branched biolubricant and reusability of [HMIm][PF_6]

The results of catalytic activity test for [HMIm][PF_6] recovery is illustrated in Fig. 1. The conversion yield and selectivity were 85.14% and 86.15%, respectively, when [HMIm][PF_6] was used for the first time (24 h). [HMIm][PF_6] remained a high catalytic activity with a conversion yield of 81.22% and selectivity of 82.10% after three recycling times. However, at the fourth round, the conversion yield and selectivity significantly decreased due to high reaction temperature and wight loss during recovery process.

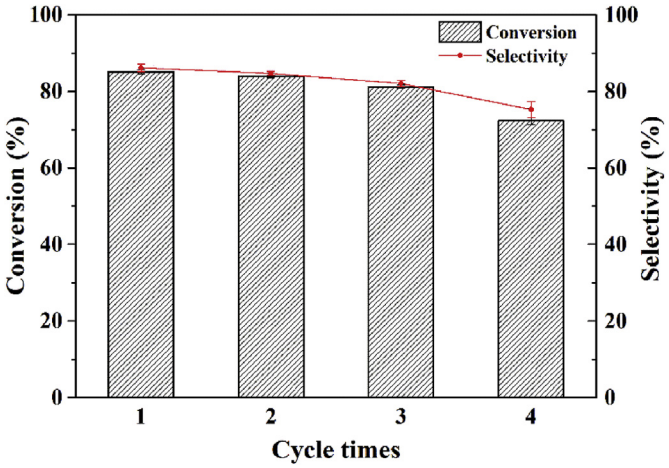


Fig. 1. Reusability of ionic liquid [HMIm][PF_6].

3.2. FT-IR and NMR spectroscopy

FT-IR and NMR spectroscopy were used to monitor the structural modification of WCO. As shown in Fig. 2, when 2-ethylhexyl esters were converted into epoxy 2-ethylhexyl esters, the characteristic peaks at 3006 cm^{-1} (the stretching of C—H next to double bond) and 1654 cm^{-1} (the stretching of double bond) evidently disappeared. The absorption peak at 826-842 cm^{-1} (the stretching of C—O—C bond of oxirane ring) was observed, which confirmed that the C═C double bonds were converted to oxirane groups by the epoxidation reaction. Besides, as plotted in Fig. 3A, the ^{1}H NMR signals at 5.45—5.25 ppm (the absorption peak of hydrogen the on double bond), 2.85—2.75 ppm, and 2.10—2.00 ppm (the absorption peaks of hydrogen on carbon next to C═C double bond) disappeared. The signals at 3.15—2.80 ppm (the absorption peaks of hydrogen on oxirane ring) were found, which also suggested that the C═C double bonds were converted to oxirane groups. Corresponding findings were confirmed in ^{13}C NMR spectra (Fig. 3B) as revealed by the vanished peaks at 128 ppm and 131 ppm (the absorption peaks of double bond carbons) and the emerged peaks at 54.2 ppm and 56.5 ppm (the absorption peaks of oxirane carbons).

When the oxirane ring was attacked by a nucleophilic reagent (octanoic acid) to form branched biolubricant, the characteristic peak at 826-842 cm^{-1} (the stretching of C—O—C bond of oxirane ring) entirely disappeared. The signals in ^{1}H NMR spectra (Fig. 3A) at 3.15—2.80 ppm (the absorption peaks of hydrogen on oxirane ring) disappeared while clear signals at 4.90 ppm (the absorption peaks of hydrogen on carbons of oxirane ring cleavage) were found.

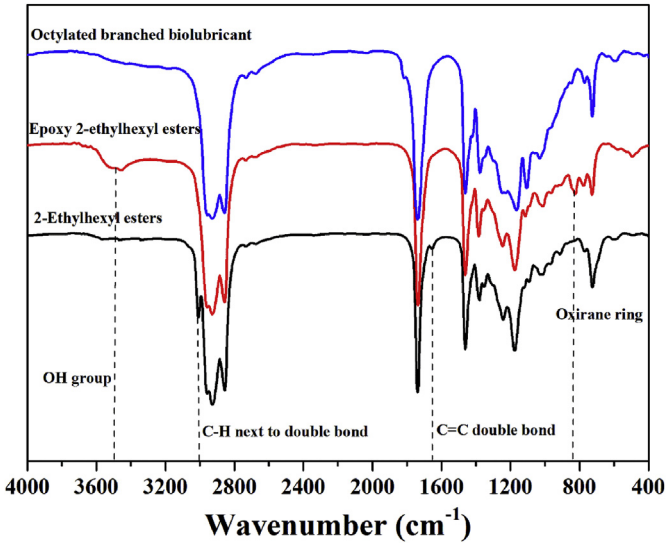


Fig. 2. FTIR spectra of products during each synthesis stage.

Furthermore, in ^{13}C NMR spectra (Fig. 3B), the absorption peaks of epoxy carbons at 54.2 ppm and 56.5 ppm vanished while the absorption peaks of carbons at branched sites after oxirane ring cracking at 73—75 ppm emerged. These results indicated that the oxirane was cleaved to form octylated branched biolubricant. It

Table 2
Fatty acid composition of waste cooking oil (WCO) by urea complexation.

Fatty acids	Molecular formula	WCO (%)[a]	Urea complexation (%)[a]
Lauric acid	C12:0	0.16 ± 0.02	0.22 ± 0.01
Myristic acid	C14:0	0.87 ± 0.06	0.46 ± 0.04
Palmitic acid	C16:0	18.92 ± 0.12	0.57 ± 0.01
Palmitoleic acid	cis-9-C16:1	1.27 ± 0.13	1.74 ± 0.05
Stearic acid	C18:0	5.75 ± 0.04	0.20 ± 0.02
Oleic acid	cis-9-C18:1	33.70 ± 0.27	36.29 ± 0.12
Linoleic acid	cis-9,12-C18:2	34.51 ± 0.12	54.47 ± 0.17
Linolenic acid	cis-9,12,15-C18:3	2.77 ± 0.01	5.54 ± 0.02
Arachidic acid	C20:0	1.11 ± 0.06	0.07 ± 0.01
Other fatty acids	—	0.78 ± 0.83	0.44 ± 0.45
Total of UFAs	—	72.25 ± 0.53	98.04 ± 0.36
Yield of non-urea complexes	61.28 ± 0.20		

[a] Mean ± SD.

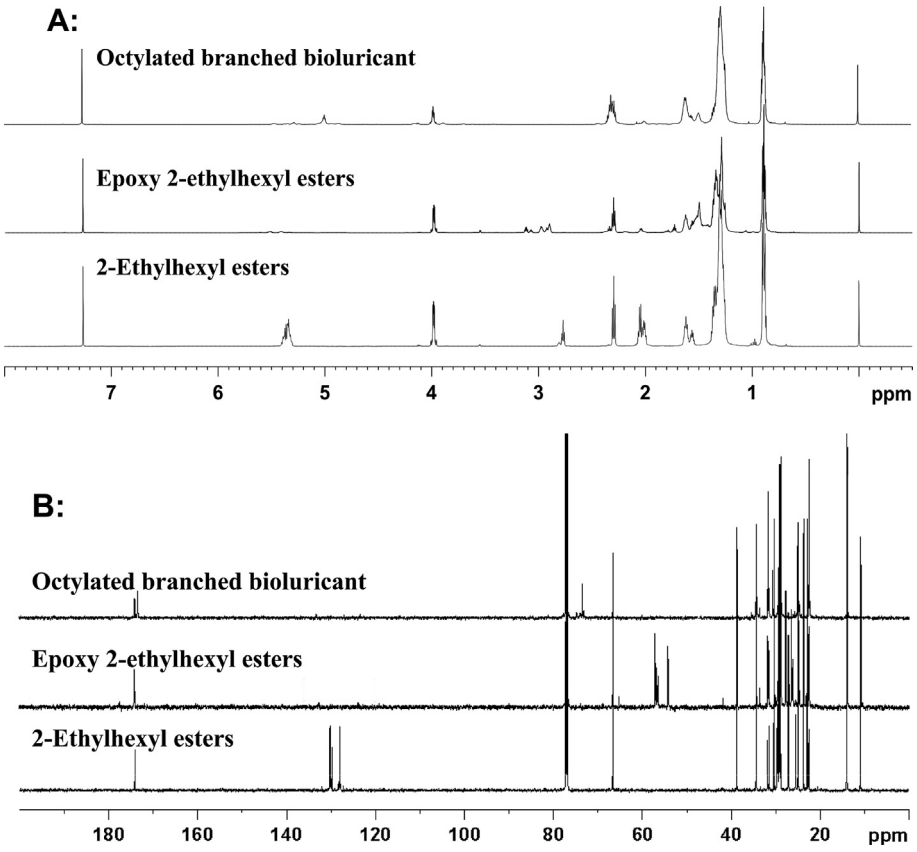


Fig. 3. ^{1}H NMR (A) and ^{13}C NMR (B) spectra of products during each synthesis stage.

should be noted that in Fig. 2, the characteristic peak of epoxy 2-ethylhexyl esters at 3500 cm^{-1} (the stretching of OH group) emerged, because the by-products were generated by oxirane ring automatic cleavage (Chua et al., 2012; Sienkiewicz and Czub, 2016).

3.3. The pour point and flash point

The main challenges for the lubricant from natural vegetable oils is the poor low-temperature performance. In this study, the low-temperature performance of modified products was evaluated through determining PP. As listed in Table 3, the PP of WCO was 2.67 °C attributed to the presence of long chain saturated fatty acids of 25% (palmitic acid and stearic acid). The low-temperature performance of 2-ethylhexyl ester was significantly improved (PP = −47.33 °C), because the content of the UFAs was higher than 98% after urea complexation. In addition, 2-ethylhexanol had a branched chain structure, which inhibited the formation of crystals (Quinchia et al., 2012; Raghunanan and Narine, 2016). However, the PP of epoxy 2-ethylhexyl esters (11.33 °C) increased rapidly with the introduction of epoxy groups. Since C=C double bonds were converted to the oxirane rings, the original *cis*-molecular conformation (curved geometry) was changed to a linear shape. Therefore, the increased intermolecular force between the ethylene oxide rings led to a more ordered accumulation of molecules and further crystallization (Borugadda and Goud, 2014). Finally, oxirane rings were opened by octanoic acid forming octylated branched biolubricant (PP = −61 °C). Due to the introduction of branches (2–4) in the middle of molecular structure, the molecules were irregularly curved and could not be packed tightly at low temperature resulting in reduced intermolecular forces and the formation of microcrystalline structure (Bouzidi et al., 2012; Raghunanan and Narine, 2016). The octylated branched biolubricant could be used as lubricant base stock for extremely cold condition.

The FP of lubricant was an important index in measuring the safety of storage and transportation. Table 3 shows that the FP of the octylated branched biolubricant was 303.67 °C. The FP increased with increase of the number of carbon atoms by epoxy and ring-opening acylation.

3.4. Rheological properties, kinematic viscosity and viscosity index

The viscosity and VI of lubricants play a critical role in lubrication performance. The oil film formed between lubricating parts and ulteriorly reduced the wear between metal parts (Campanella et al., 2010). The viscosity and VI of obtained products during WCO modification were listed in Table 3. When UFAs esters were modified via epoxidation and further acylation, the molecular weight increased gradually and the corresponding kinematic viscosity also raised. However, the trend of VI showed a decrease and then a certain recovery through epoxidation and acylation of octanoic acid (McNutt and He, 2016). This attributed to polar functional groups and the van der Waals force between molecules increasing with high degree of opening and branching (Valle et al., 2018; Kulkarni et al., 2013). VI of 149 and higher kinematic viscosity (40 °C = 67.51 cSt and 100 °C = 10.74 cSt, respectively) of octylated branched biolubricant presented an excellent viscosity-temperature performance and guaranteed sufficient oil film thickness in lubrication process. Besides, the rheological behaviors of WCO and octylated branched biolubricant were shown in Fig. 4a and b. The linear relationship between shear stress and shear rate

Table 3
Physico-chemical properties of waste cooking oil (WCO) and octylated branched biolubricant.

Physico-chemical properties	WCO	2-Ethylhexyl esters	Epoxy 2-ethylhexyl esters	Octylated branched biolubricant
Acid value (mgKOH/g)[a]	67.82 ± 0.12	0.49 ± 0.08	8.22 ± 0.15	0.55 ± 0.11
Iodine value (gI_2/100 g)[a]	98.14 ± 0.64	101.83 ± 0.41	7.42 ± 0.13	2.50 ± 0.54
Oxirane content (Experimental, mass%)[a]	–	–	5.37 ± 0.06	0.06 ± 0.01
Oxirane content (Theoretical, mass%)	–	–	6.03	–
Relative conversion (%)[a]	–	–	89.05 ± 0.99	98.88 ± 0.56
Kinematic viscosity at 40 °C (cSt)[a]	34.33 ± 0.78	8.43 ± 0.27	21.76 ± 0.18	67.51 ± 0.20
Kinematic viscosity at 100 °C (cSt)[a]	7.36 ± 0.08	2.83 ± 0.05	4.71 ± 0.14	10.74 ± 0.05
Viscosity index	188	216	140	149
Pour point (°C)[a]	2.67 ± 0.47	−47.33 ± 1.79	11.33 ± 1.41	−61.00 ± 0.43
Flash point (°C)[a]	177.33 ± 2.45	213.33 ± 4.32	232.00 ± 6.69	303.67 ± 7.02
Oxidative induction time (min)	12	–	–	21

[a] Mean ± SD.

all were excellent at different temperature (30, 40, 50, 60, 80 and 100 °C), showing a superior Newtonian fluidity. Fig. 4.c showed a decrease of kinematic viscosity of WCO and octylated branched biolubricant with the increase of temperature, which was mainly caused by the thermal movement between molecules (Borugadda and Goud, 2019).

3.5. Thermal-oxidation stability analysis

The thermal-oxidation stability of lubricants presents the ability of the molecule to resist decomposition and the tendency of antioxidant formation at high temperature. Lubricants with low thermal-oxidation stability are prone to be fractured and rapidly oxidated. Self-polymerization between lubricant molecules results in a higher viscosity and a loss of lubrication properties (Majano et al., 2011). TG analysis and RPVOT were conducted to evaluate the thermal-oxidation stability of WCO and octylated branched biolubricant. Fig. 5 showed that the weight loss process of WCO and octylated branched biolubricant in N_2 atmosphere and air atmosphere, and the specific data was recorded in Table 4. In N_2 and air atmosphere, the octylated branched biolubricant had a single-stage degradation mode. However, WCO had a multi-level degradation model, causing by the mixtures of free fatty acids and glycerides. In N_2 atmosphere (Fig. 5a), the onset temperatures of WCO and octylated branched biolubricant were 221.67 °C and 326.35 °C, respectively. The thermal stability of octylated branched biolubricant was significantly improved due to the disappearance of unsaturated sites and the increase of branching degree (Chan et al., 2018). According to differential thermogravimetry (DTG) curves, the weight loss of octylated branched biolubricant at 250–450 °C was attributed to the evaporation and decomposition of hydrocarbons, while the weight loss of WCO at 180–320 °C was due to the evaporation of free fatty acids. The weight loss between 320 and 500 °C was due to the evaporation and decomposition of glycerides (Valle et al., 2018).

Similarly, the oxidation stability of WCO and octylated branched biolubricant were detected in air atmosphere (Fig. 5b). The oxidative onset temperatures of WCO and octylated branched biolubricant were 213.45 °C and 312.06 °C, respectively, which showed that the oxidation stability of the octylated branched biolubricant was improved. Because the easily oxidized sites, C=C double bonds, were replaced by the acylated branch chains. Therefore, the formation of oxidative free radicals was avoided. Interestingly, the weight loss of WCO in air atmosphere was divided into three stages. Free fatty acids were evaporated between 150 and 300 °C. Glycerides began to be evaporated and decomposed at 300–480 °C. The final weight loss stage at 480 °C above was owing to the further combustion and volatilization of low molecular hydrocarbons, carbon dioxide and carbon monoxide (Borugadda and Goud, 2014). Furthermore, by means of RPVOT (Table 3), the oxidative induction time of octylated branched biolubricant (21 min) was expanded compared to WCO (12 min), which also confirmed that the oxidation stability of the octylated branched biolubricant was prominently improved.

3.6. Tribological performance

The lubrication performance of WCO and octylated branched biolubricant were evaluated comparing with a mineral based lubricant with the same viscosity grade through HFRR, and the specific data was recorded in Table 5. The COF and WSD were typically used to evaluate the lubrication performance and the results were shown in Fig. 6. During the process of friction test, the COF of octylated branched biolubricant was steady at 0.09, while the COF of WCO fluctuated between 0.06 and 0.09. The anti-friction performance of WCO and octylated branched biolubricant were inferior to that of mineral based lubricant with the same viscosity grade (WSD = 343 μm, COF = 0.15). Due to the existence of polar functional groups, such as carboxyl groups, ester groups and hydroxyl groups in WCO and octylated branched biolubricant, the lubricants molecules between metal parts were tightly adsorbed on metal surface to form a stable lubricating film, which reduced direct friction between metal parts (Borugadda and Goud, 2019; Chan et al., 2018). In addition, the COF of WCO was lower than that of octylated branched biolubricant (0.07 *vs* 0.09), but the WSD was opposited (345 μm *vs* 203 μm). The main reason was that there were free fatty acids in WCO and the carboxyl groups had corrosion action on the metal surface (Salimon et al., 2012b), and more viscous octylated branched biolubricant (67.51 cSt in 40 °C) can provide a thicker oil film to ensure that the friction parts were covered as much as possible (Film coverage rate was 90%).

To compare the performance of octylated branched biolubricant in this work with other biolubricants obtained from WCO, we have added Table 6 to the manuscript, also including chemically modified strategies and catalysts. A higher kinematic viscosity of the synthesized octylated branched biolubricant led to a thicker oil film and superior lubrication properties when working between metal parts. After urea complexation, the content of UFAS increased, providing more branch positions for subsequent ring opening, so the PP of octylated branch biolubricant was as low as −61 °C. Oxidative and thermal stabilities of the octylated branched biolubricant were significantly improved due to the disappearance of C=C bonds after epoxidation and ring opening. Meanwhile, lipases and [HMIm][PF_6] had high catalytic efficiency, high specificity, good

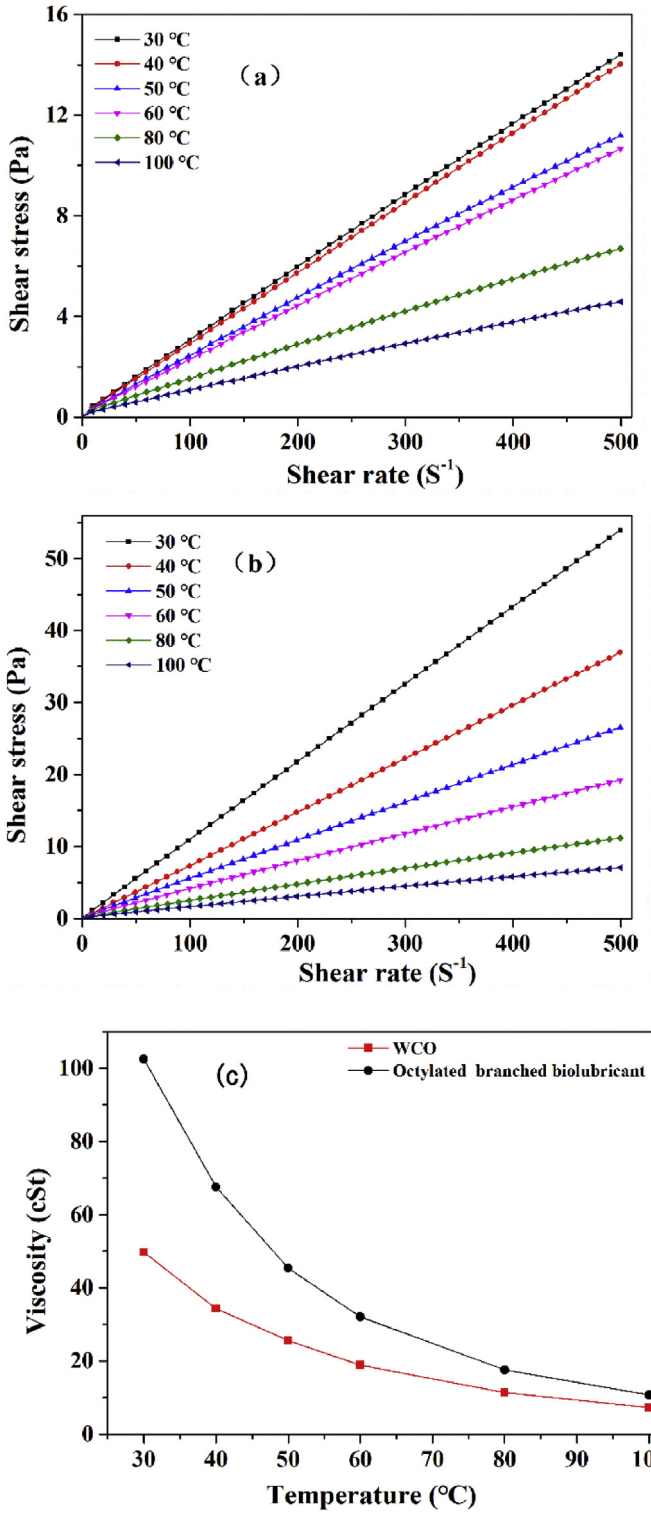


Fig. 4. Rheological behaviour of waste cooking oil (WCO) and octylated branched biolubricant. (a) Shear rate and shear stress relation of WCO at various temperatures; (b) Shear rate and shear stress relation of octylated branched biolubricant at various temperatures; (c) Temperature and viscosity relation of WCO and octylated branched biolubricant at various temperatures.

recyclability in chemically structural modification of UFAs, [HMIm][PF_6] was reused for 3 times keeping high catalytic activity. Although Borugadda and Goud (2019) prepared biolubricants with low PPs (−27.15 °C and −28.45 °C), improved thermal-oxidation

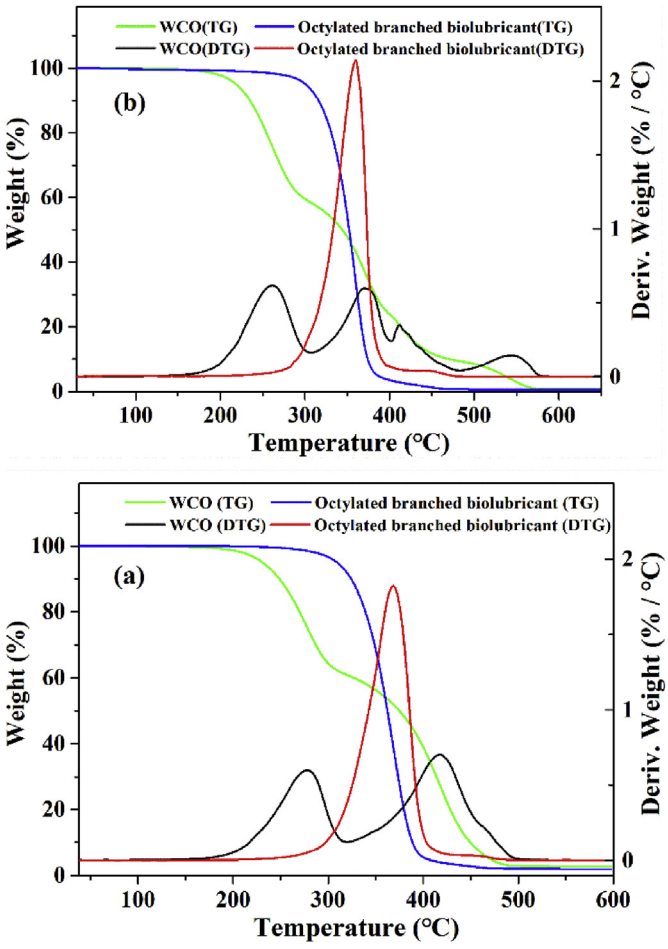


Fig. 5. TG curves of waste cooking oil (WCO) and octylated branched biolubricant in nitrogen (a) and air atmosphere (b).

stability and good lubricity, the kinematic viscosity (40 °C = 10.57 or 8.33 cSt) was not sufficient to provide film thickness for lubrication.

4. Conclusion and future prospects

In this study, it was reported a green and efficient strategy for producing branched biolubricant from WCO catalyzed by lipases and [HMIm][PF_6] which had high catalytic efficiency, high specificity, simplified separation and considerable recyclability. WCO was hydrolyzed by *Candida* sp. *99–125* lipase and 98.04% of UFAs was collected by urea complexation. Subsequently, the esterification of UFAs with branched alcohol (2-ethylhexanol) by the lipase (Novozym 435) directly produced new branched esters and further epoxidated. Finally, the epoxy groups were attacked by a cheap nucleophilic reagent, octanoic acid, to obtain the octylated branched biolubricant catalyzed by [HMIm][PF_6]. [HMIm][PF_6] maintained high catalytic activity with a conversion of 81.22% and selectivity of 82.10% after three recycling times. The obtained octylated branched biolubricant showed excellent lubricant properties, such as low PP of −61 °C, high VI of 149, and high thermal-oxidation stability (onset temperature of 326.35 °C, oxidative onset temperature of 312.06 °C). HRFF results demonstrated that its lubrication performances (COF of 0.09, WSD of 203 μm) were much better than that of mineral based lubricant with the same viscosity grade. Biolubricants will play an increasing role of the lubricant

Table 4
Thermal-oxidation stability of waste cooking oil (WCO) and octylated branched biolubricant.

Samples	Thermal stability (°C)			Oxidative thermal stability (°C)				
	OT[a]	SMT_1[b]	SMT_2	OOT[c]	SMT_1	SMT_2	SMT_3	SMT_4
WCO	221.67	277.93	417.70	213.45	261.10	370.46	411.90	544.39
Octylated branched biolubricant	326.35	368.28		312.06	360.11			

[a] OT: onset temperature.
[b] SMT: signal maximum temperature.
[c] OOT: oxidative onset temperature.

Table 5
The tribological properties of waste cooking oil (WCO) and octylated branched biolubricant.

Samples	Wear spot diameter (μm)	Coefficient of friction	Film coverage rate
Mineral oil	343	0.15	17%
WCO	345	0.07	9%
Modified lubricant	203	0.09	90%

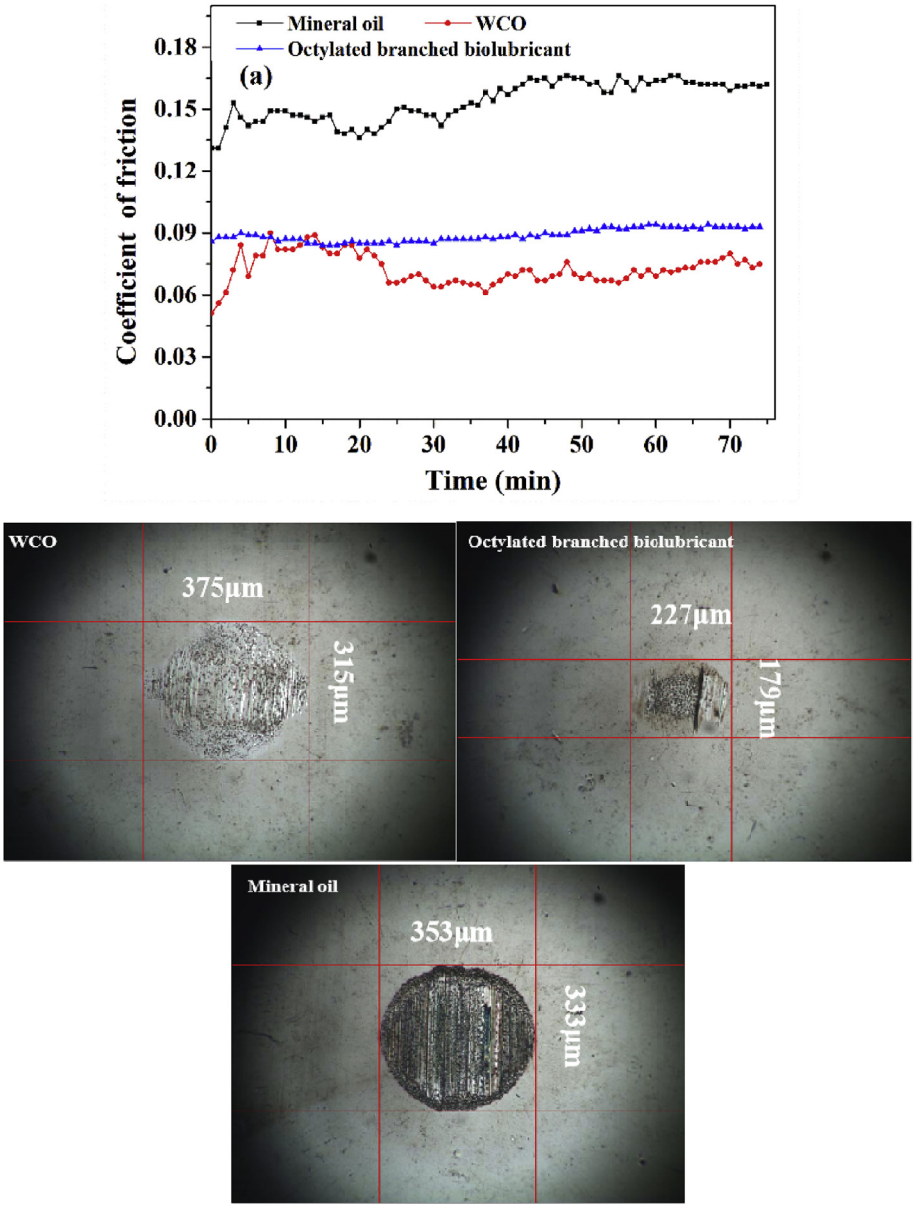


Fig. 6. (a) Variation in coefficient of friction of waste cooking oil (WCO), octylated branched biolubricant and a mineral oil. (b) Corresponding electron micrographs of wear scars.

Table 6
Comparison of different chemical modification of waste cooking oil (WCO) to obtain biolubricants.

Reactants	Catalyst	Modification methods	Catalyst reusability	Products	Viscosity 40 °C (cSt)	Pour point (°C)	Oxidative/thermal stability	Lubrication performance		Ref.
								Coefficient of friction	Wear spot diameter (μm)	
WCO, methanol, trimethylolpropane	CaO	Two steps transesterification	–	TMP triester	30	–	–	–	–	Ghafar et al. (2019)
WCO, methanol, trimethylolpropane	KOH and potassium carbonate	Transesterification and esterification	–	TFATE	15.5	−9	–	–	–	Soufi et al. (2019)
Waste cooking oil methyl esters (WCOME), acetic acid,	IR-120	Epoxidation	–	Epoxidized WCOME	10.42	10	Onset temperature:187 °C, oxidative onset temperature:245.5 °C	–	–	Borugadda and Goud (2016)
WCO, methanol, isooctanol, isotridecanol and isooctadecanol	Amberlyst 15 and CaO	Epoxidation, two-step transesterification	–	Three epoxidized branched esters	15.9 −43.4	(-24) to (−17)	Oxidative onset temperature:163.9 −194.5 °C	0.05−0.07	510−650	Li and Wang (2015)
Epoxidized WCO/ epoxidized WCOME, 2-ethylhexanol, hexanoic anhydride	H_2SO_4, IR-120 and Amberlyst 15	Epoxidation, hydroxylation and hexanoylation	–	Hexanoylated WCO/ hexanoylated WCOME	10.57/ 8.33	−28.45/-27.15	Onset temperature:286.33/ 293 °C, oxidative onset temperature:230.7/ 247.5 °C	0.08/0.1	232/209	Borugadda and Goud (2019)
WCO, 2-ethylhexanol, octoic acid	Lipase and ionic liquid	Enzymatic hydrolysis and urea complexation, enzymatic esterification and epoxidation, ring opening and acylation	3 times (ionic liquid)	Octylated branched biolubricant	67.51	−61	Onset temperature: 326.35 °C, oxidative onset temperature: 312.06 °C	0.09	203	This work

market in the near future. This work provided an available scheme for the production of high-performance biolubricants from WCO or plant oils containing high-content polyunsaturated fatty acids. Ionic liquids with simplified separation and high catalytic activity also had great potential in chemically structural modification of biolubricant.

CRediT authorship contribution statement

Wei Zhang: Investigation, Data curation, Writing - original draft. **Hairui Ji:** Writing - review & editing. **Sen Ma:** Formal analysis. **Wei Xiong:** Formal analysis. **Changjing Chen:** Methodology. **Biqiang Chen:** Supervision, Investigation. **Xu Zhang:** Supervision, Investigation.

Declaration of competing interest

No conflict of interest exists in the submission of this manuscript, and the manuscript is approved by all authors for publication. I would like to declare on behalf of my co-authors that the work described was original research that has not been published previously, and not under consideration for publication elsewhere, in whole or in part. All the authors listed have approved the manuscript that is enclosed.

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